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SUMMARY

Applied AI engineer and AI FDE who works embedded with users to turn ambiguous problems into reliable, adopted systems. I design, evaluate, deploy, and roll out agentic AI, RAG/GraphRAG, and ML systems, with outcomes including a week-long workflow reduced to under two minutes, token usage and associated costs reduced by 76% to 99%, and systems supporting hundreds of colleagues in regulated environments.

EXPERIENCE

Department for Education, HEO Data Scientist (GORS), Jan 2024 to Present

- **Designed and delivered** a Benefits Realisation Agent for PMO colleagues, translating workshop language into schema-constrained extraction, bounded generation, tool calls, validation, and retries. Reduced a routine that previously took about a week to around 90 seconds end to end; it now supports every project overseen by the division and produces documents with 95%+ accuracy.
- **Architected** a FastAPI GraphRAG knowledge base with extended functionality that makes it queryable by other agents and tools such as Claude Code, Codex, and GitHub Copilot CLI. Community detection makes semantic and lexical relationships discoverable across a corpus, while community, sub-community, and theme retrieval reduces token usage and associated costs by 76% to 99%.
- **Architected and led** a cross-directorate analytics platform, replacing three legacy tools and supporting roughly 300 operations and policy colleagues. Reduced time to report production from a two-week average to under five minutes for a defined output set, with selected outputs knitted into a slide deck in under one minute.
- **Rebuilt** a confidential production risk-assessment system in Python and Databricks, adding MLflow, feature-store inputs, calibrated XGBoost, drift monitoring, and Azure DevOps CI/CD; it now supports 60 colleagues conducting individual investigations in their daily operations.
- **Migrated and operationalised** a budget-forecasting model on Databricks in 11 working days against a one-month window, replacing its orchestration layer, moving inputs to Unity Catalog, optimising PySpark ETL, and building its MLOps layer to preserve business continuity.

Aramayo Lab, Texas A&M University, AI Consultant, Jan 2023 to Present

- **Built and continue to develop** a Scientific Workbench for genomics and bioinformatics workflows, connecting research data, literature, tools, and compute. Three labs are currently engaged, with expansion toward 12 by year-end; task evaluation is ongoing.
- **Fine-tuned and evaluated** small task-bound language models using paired runs, sham-control arms, and repeated-sampling consistency tests; integrated role-based local and hosted model routing with university HPC dispatch so tasks run in an appropriate environment.

OPEN-SOURCE AND INDEPENDENT ENGINEERING

Author and maintainer of Nuthatch, a knowledge-base GraphRAG system; Testudo, a secure containerised runtime; and Local Model Workforce, a fine-tuned specialist LLM roster for local deployment. I integrate these public tools into the Scientific Workbench for orchestration, retrieval, sandboxed tool use, and model evaluation across complex scientific workflows.

SKILLS

- **AI systems:** OpenAI Codex, Claude Code, Pi agent harness, MCP server authoring, context and memory management extensions, LLM orchestration, LangChain, LangGraph, agentic workflows, RAG/GraphRAG, schema-constrained extraction, tool use, LoRA fine-tuning, and local inference.
- **Production engineering:** Python, TypeScript, FastAPI, scikit-learn, PyTorch, Databricks, MLflow, Spark/PySpark, PostgreSQL, SQL, Docker, Azure DevOps CI/CD, Plotly Dash, Electron, Node.js, React, Vite, model monitoring, and auditable workflows.
- **Delivery:** user discovery, requirements, technical scoping, stakeholder communication, adoption, workshops, and mentoring.

EDUCATION

- Texas A&M University, PhD Candidate, Computational Genomics (Spring 2027).
- University College London, MSci Cell and Developmental Biology (2:1), 2016.